

TABLE V
Heated Tweezers from Different Vendors

Specifications	Manufacturer				
	OK International	HAKKO	Srinivasa Electronics	Metal	Weller
Model	MFR-1140	Hakko FM-2022	PLUTO 555	MFR-1140	WMRT
Power consumption (W)	70	140	80	70	2×40
Temperature range (°C)	Smarteat-based variable power system	—	180-350	—	50-450
Tip-to-ground resistance (Ω)	< 2	< 2	< 2	< 2	—
Tip-to-ground potential (mV)	< 2	< 2	0.2	< 2	—
Heating element	Patented Smarteat-based heater	Composite heater	Ceramic type	—	—
Cord length (m)	1.22	1.2	1.2	1.83	1 (approx.)
Dimensions (mm)	90×142×197	length - 148	—	90×142×197	—
Weight (gm)	120	64	2000	2300	—
ESD safe (Yes/No)	Yes	Yes	Optional	Yes	Yes



Fig. 3: A vacuum pump-type de-soldering station

tion of the melted solder from PCB. The heated nozzle is put over the joint with its pin at the centre to melt the solder and, once the solder has melted, a button is pressed to perform suction of the melted solder. Such tools can de-solder almost all types of components, except the ball grid array (BGA) types. BGA package needs a different de-soldering tool specifically designed for the purpose.

Table III shows some vacuum pump de-soldering stations from different vendors and Table IV shows some hot-air rework systems from different vendors.

Heated tweezers have more targeted heat transfer than the hot-air rework stations. They provide a fast and efficient method to solder and de-solder SMDs such as chip resistors, chip capacitors, SOTs, flat packs and DIP



Fig. 4: Hot air rework station

ICs. Table V lists some heated tweezers available from different vendors.

Another important area is to carefully select various hand tools. There are several manufacturers around the globe producing high-quality electronics tool-kits with all sorts of tools that you would ever need in your electronics career. There are also some suppliers who take the best parts from different manufacturers and assemble them to make a single kit. But you can skip on buying the complete tool kit as it will be expensive, besides having several tools that you would never use in your life. We suggest that, to begin with, you buy only a PCB holder, needle nose plier, a set of tweezers, smart tweezers (optional), snips and clippers, wire strippers,



Fig. 5: Heated tweezers

screwdriver set, crimping tool and IC extractor. The other items can be bought later, if needed.

The last thing to look at is the magnifier. Selection of the magnifier will depend on your requirement. Interestingly in most cases, you can do without high-magnification sophisticated magnifiers. The regular magnifier is easily available such as triplet loupe, clamp stand with magnifier, three lens multi-magnifier, etc. If you are working for long durations, you can consider buying one of the head-mounted magnifiers that has three lenses: one binocular lens, a second binocular lens behind the first that flips up and a pivoting loupe. You can use different combinations of these lenses to achieve magnifications like 1.8X, 2.3X, 3.7X and 4.8X.

Only in special cases, such as working with fine-pitch ICs, you would need more advanced stereoscopic microscope which is expensive. So consider buying one only when you really need it. ●

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